

Weighted convolution of double series

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Abstract

The Cauchy product of two double power series, on coefficient arrays $\mathbb{N}^2 \rightarrow \mathbb{R}$: $(f * g)_k = \sum_{a \leq k} f_a g_{k-a}$, with the weighted ℓ^1 norm $N_\rho(f) = \sum_k |f_k| \rho^{|k|}$ and weighted summability. The two results (Astra #7(a)) are the closure $N_\rho(f * g) \leq N_\rho(f) N_\rho(g)$ (with weighted summability of the product) and multiplicativity of evaluation $\sum_k (f * g)_k u^i v^j = (\sum_k f_k u^i v^j)(\sum_k g_k u^i v^j)$ for $|u|, |v| \leq \rho$. Both come from one finite-fibre bridge: the fibre of the addition map $(\mathbb{N}^2)^2 \rightarrow \mathbb{N}^2$ over k is in bijection with the box $\{a \leq k\}$, so `HasSum.tsum_fiberwise` regroupes the product family. No `HasAntidiagonal` instance on $\mathbb{N}^2$ is needed. Module 1 of the coefficient closure for $\eta e^{\beta s \xi}$. Zero sorry/axiom.

1 The things formalised

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def psub (k a :  $\mathbb{N} \times \mathbb{N}$ ) :  $\mathbb{N} \times \mathbb{N}$  := (k.1 - a.1, k.2 - a.2)
def box (k :  $\mathbb{N} \times \mathbb{N}$ ) : Finset ( $\mathbb{N} \times \mathbb{N}$ ) := Finset.range (k.1 + 1)  $\times$  Finset.range (k.2 + 1)
noncomputable def conv (f g :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (k :  $\mathbb{N} \times \mathbb{N}$ ) :  $\mathbb{R}$  :=  $\sum_{a \in \text{box } k} f a * g (\text{psub } k a)$ 
def WSummable (f :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) : Prop := Summable fun k => |f k| *  $\rho^{k.1 + k.2}$ 
noncomputable def wnorm (f :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) :  $\mathbb{R}$  :=  $\sum_k |f k| * \rho^{k.1 + k.2}$ 
noncomputable def fiberEquiv (k :  $\mathbb{N} \times \mathbb{N}$ ) : (box k)  $\rightarrow$  addFiber k
theorem fiber_tsum_eq (H : ( $\mathbb{N} \times \mathbb{N}$ )  $\times$  ( $\mathbb{N} \times \mathbb{N}$ )  $\rightarrow$   $\mathbb{R}$ ) (k :  $\mathbb{N} \times \mathbb{N}$ ) :
  ' p : addFiber k, H p =  $\sum_{a \in \text{box } k} H (a, \text{psub } k a)$ 
theorem wnorm_conv_le (h : 0 <  $\rho$ ) (f g :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (hf : WSummable f)
  (hg : WSummable g) :
  WSummable (conv f g) wnorm (conv f g) wnorm f * wnorm g
theorem dblSum_abs_le_wnorm (u v :  $\mathbb{N} \times \mathbb{N}$ ) (f :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (hf : WSummable f)
  (hu : |u| <  $\rho$ ) (hv : |v| <  $\rho$ ) :
  (Summable fun k :  $\mathbb{N} \times \mathbb{N}$  => ||f k * uk.1 * vk.2||)
  ' k :  $\mathbb{N} \times \mathbb{N}$ , ||f k * uk.1 * vk.2|| wnorm f
theorem dblSum_conv (u v :  $\mathbb{N} \times \mathbb{N}$ ) (f g :  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{R}$ ) (hf : WSummable f) (hg : WSummable g)
  (hu : |u| <  $\rho$ ) (hv : |v| <  $\rho$ ) :
  dblSum (conv f g) u v = dblSum f u v * dblSum g u v
```

2 Mathematics

The weighted product family $H(a, b) = (|f_a| \rho^{|a|})(|g_b| \rho^{|b|})$ is absolutely summable with sum $N_\rho(f) N_\rho(g)$ (`summable_mul_of_summable_norm`, `tsum_mul_tsum_of_summable_norm`). Grouping it along the fibres of addition (`HasSum.tsum_fiberwise`) and converting each fibre sum to the finite sum over the box gives $B_k = \rho^{|k|} \sum_{a \leq k} |f_a| |g_{k-a}|$ with $\sum_k B_k = N_\rho(f) N_\rho(g)$, and $|(f * g)_k| \rho^{|k|} \leq B_k$ by the triangle inequality; the weight identity is $\rho^{|a|} \rho^{|k-a|} = \rho^{|k|}$. Evaluation multiplicativity is the same grouping applied to the signed family $(f_a u^{a_1} v^{a_2})(g_b u^{b_1} v^{b_2})$, absolutely summable because

$|u^i v^j| \leq \rho^{i+j}$, with $u^{a_1} u^{(k-a)_1} = u^{k_1}$ on each fibre. As Astra #7 stressed, the weighted family is grouped, never the unweighted one: weighted summability at $\rho < 1$ does not give unweighted summability.

3 Paper-facing meaning

This is the algebra needed to build the coefficient array of $\eta(u, v)e^{\beta s \xi(u, v)}$ from the double series of ξ and η (next units: convolution powers, the exponential series and its majorant, then the amplitude adapter feeding `twoD_cutoff_analytic`), so that the analytic $d = 2$ cutoff theorem applies to the chart amplitude of `thm:TaylorTree` under the paper's analyticity hypothesis on ξ, η alone.

4 Lean notes

- Avoid antidiagonals on $\mathbb{N}^2$ altogether: index the convolution by the box $\{a \leq k\}$ with componentwise truncated subtraction, and prove one `Equiv` between the box and the addition fibre; `Equiv.tsum_eq` plus `Finset.tsum_subtype` then give the fibre-sum bridge.
- After `rw [← e.tsum_eq]` the summand is `H (e c)`; a `change` to the unfolded pair is needed before `Finset.tsum_subtype` matches.
- For a subtype of a product, `p.1` is the value, not its first component; write `(p : ( × ) × ( × )).1`.
- `Real.norm_of_nonneg` with an explicit `positivity` proof converts norms of weighted absolute values; `simp` with `Real.norm_eq_abs` tends to reassociate.
- Rewriting `k` via `a + psub k a = k` also rewrites inside `psub k a`; derive component identities with `omega` from `mem_box` instead.